Riding the Wave?

Who Starts, Follows and Gains from Machine Learning Research Trends

A Study of 54,579 ICLR Submissions, 2017–2026

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Outline

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Motivation: Research Fashions in Machine Learning

Share of ICLR submissions that ...	2017	2026
propose a new architecture	24%	8%
build on a foundation model	0.3%	54%
carry a branded title ("NAME: subtitle")	9%	41%

Are these fashions? What do they displace, who drives them, and are followers rewarded?

Research Questions and Hypotheses

Research question	Key prediction
RQ1 Rise: what does a fashion displace?	More foundation-model (FM) papers in a topic $\rightarrow$ fewer new-architecture papers (H1d)
RQ2 Following: who introduces new work?	Newcomers imitate: fewer new-method papers, more benchmark papers (H2b, H2c)
RQ3 Reward: are followers rewarded?	Crowded topics $\rightarrow$ lower acceptance (H3a)

All predictions were written down before any test was run (10 hypotheses in total).

Related Work and Positioning

Who drives new work? Two views

- *Fashion and herding:* newcomers imitate (Abrahamson, 1991; Bikhchandani et al., 1992)
- *Outsider innovation:* newcomers bring new ideas (Foster et al., 2015; Azoulay et al., 2019)

What we add to meta-research on machine learning

(Koch et al., 2021; Birhane et al., 2022)

- All submissions, not only accepted papers
- Topics defined without hindsight
- Tests registered in advance (Nosek et al., 2018), plus a held-out replication

Data and Measurement

Construct	Measure
Papers	43,480 ICLR submissions, 2017–2026 (accepted, rejected and withdrawn); 11,099 held-out papers for the replication
Topic	SPECTER2 embeddings of title + abstract (Singh et al., 2023), clustered into $k = 100$ topics; results agree at $k = 75$ and 150
Contribution type	LLM with a fixed codebook (method, benchmark, empirical, theory, other); blind second pass $\kappa = 0.93$
Newcomer paper	No author has an earlier ICLR submission in the topic (73% of papers)
Acceptance	Final decision; locked away until the single registered run

Hindsight Bias in Topic Maps

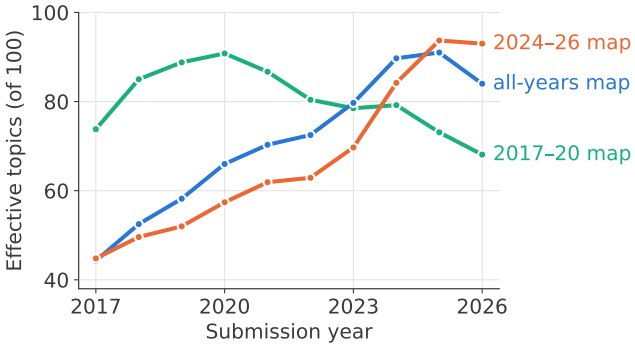


Figure 1: Topic diversity per year under three topic maps.

Topic map

The 100 topic clusters, fitted on papers from chosen years; every paper is then assigned to its nearest topic.

Each map finds peak diversity in the years it was fitted on.

⇒ The fitting window shapes the history one sees.

Consequence: Conclusions Depend on the Map

- Emerging topics ($\geq 3\times$ growth): **8** on the 2017–20 map vs. **29** on the all-years map.
- Do topics age, from new methods towards benchmarks? All-years map: no. 2017–20 map: yes.

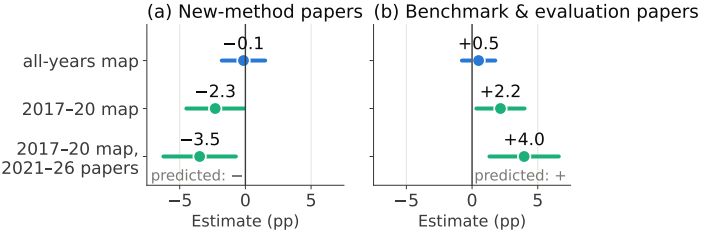


Figure 2: Same papers, same model: coefficient on topic maturity, 95% CI (2017–20 rows exploratory).

$\Rightarrow$ Every registered result must also hold on the 2017–20 map and a year-balanced map.

Empirical Strategy

Model (linear probability)

$$y_i \;=\; \beta\, x_i \;+\; \mu_{\text{topic}(i)} \;+\; \tau_{\text{year}(i)} \;+\; \varepsilon_i$$

- y_i: paper type or acceptance (0/1); x_i: e.g. newcomer paper
- Papers compared within the same topic and year; standard errors clustered by topic

Supported only if

1. predicted sign, Holm-adjusted $p < 0.05$
2. same sign in $\geq 75\%$ of alternative specifications
3. same sign on both hindsight checks

Protocol

Registered in advance, acceptance locked, code frozen, run once; held-out tests registered before their run.

RQ1: Where Foundation Models Spread, New Architectures Fade

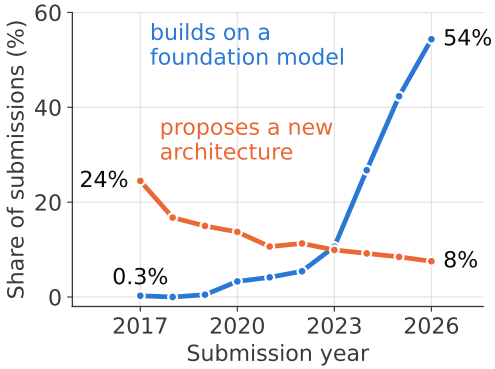


Figure 3: Field-wide trends (descriptive).

Within topic and year, FM adoption 0 → 100%:

New-architecture papers	Discovery	Held-out
Change (pp)	-8.0	-11.0
95% CI	$[-12.1, -3.8]$	$[-18.1, -3.9]$

Supported and replicated

Topics at 50% vs. 10% FM adoption: about one third fewer new-architecture papers.

RQ2: Newcomers Bring New Methods, Opposite to Our Prediction

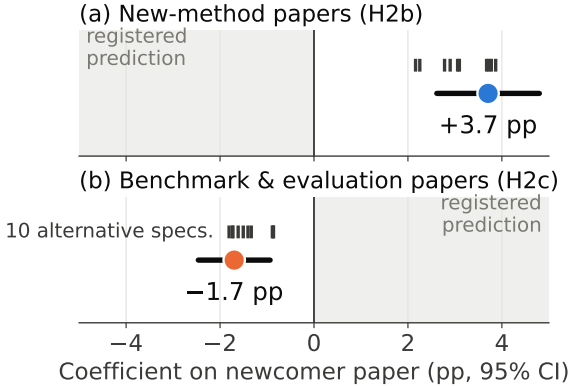


Figure 4: Discovery; within topic and year; grey: predicted side.

Newcomer paper (pp)	Method	Benchmark
Predicted	−	+
Discovery	+3.7	−1.7
Held-out	+3.6	−2.1

Reversed predictions registered, then **replicated** on held-out papers.

Reading

Consistent with outsider innovation, not with herding.

RQ3: Is Following the Fashion Rewarded?

Association with acceptance (pp, 95% CI)	Discovery	Held-out
Crowded topic, per log unit of prior papers (H3a)	-1.7 $[-3.7, +0.3]$	-1.5 $[-5.2, +2.2]$
Branded title (exploratory)	no premium	–
New architecture $\times$ topic FM adoption (exploratory)	$+8.3$ $[+1.3, +15.4]$	–

Table 1: Within topic and year. Crowding: neither split could detect effects below about 4–5 pp.

Key point

No detectable reward for following the fashion; papers that go against the tide may even do better (exploratory).

Do the Results Hold on Held-out Papers?

Three of four key results replicated; crowding stays inconclusive.

Hypothesis	Pred.	Discovery	Held-out [95% CI]	Verdict
H1d FM adoption → new-architecture papers	−	−8.0	−11.0 [−18.1, −3.9]	supported
H2b′ newcomer → new-method paper	+	+3.7	+3.6 [+1.5, +5.7]	supported
H2c′ newcomer → benchmark paper	−	−1.7	−2.1 [−3.3, −0.8]	supported
H3a crowded topic → acceptance	−	−1.7	−1.5 [−5.2, +2.2]	inconclusive†

Table 2: 11,099 papers from a released former test split, never analysed before one registered run: an out-of-sample replication, not a certified untouched holdout. Supported: one-sided Holm $p < 0.05$; pp. † Too weak to tell (below MDE_{80}).

Also registered: post hoc tests from discovery, e.g. far jumps from a team's own past work go with lower acceptance (replicated). Full table in the report.

Contributions

1. **Method:** topic maps fitted with hindsight can manufacture trends; an early-year map is a cheap check.
2. **Design:** registered, outcome-blind, single-run protocol, with a held-out replication.
3. **Findings** (associations):
   - where foundation models spread, new architectures fade (replicated)
   - newcomers lean towards new methods, not imitation (replicated)
   - no detectable acceptance reward for following the fashion

Limitations and Future Work

Limitations

- One venue; associations, not causes
- LLM labels: consistent ($\kappa = 0.93$), not yet human-validated
- Held-out papers: a released former split, not certified untouched
- 2021 author-ID change undercounts incumbents

Future work

- A prospective test on future ICLR submissions
- ACL Anthology / ARR
- Human validation of contribution labels

Thank you

Questions and comments welcome

Many thanks to my supervisor, A/Prof Jianzhong Qi, for guidance and feedback.

References

Abrahamson, E. (1991). Managerial fads and fashions: The diffusion and rejection of innovations. *Academy of Management Review*, 16(3), 586–612.

Azoulay, P., Fons-Rosen, C., & Graff Zivin, J. S. (2019). Does science advance one funeral at a time? *American Economic Review*, 109(8), 2889–2920.

Bikhchandani, S., Hirshleifer, D., & Welch, I. (1992). A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5), 992–1026.

Birhane, A., Kalluri, P., Card, D., Agnew, W., Dotan, R., & Bao, M. (2022). The values encoded in machine learning research. In *Proceedings of ACM FAccT*.

Foster, J. G., Rzhetsky, A., & Evans, J. A. (2015). Tradition and innovation in scientists' research strategies. *American Sociological Review*, 80(5), 875–908.

Koch, B., Denton, E., Hanna, A., & Foster, J. G. (2021). Reduced, reused and recycled: The life of a dataset in machine learning research. In *NeurIPS Datasets and Benchmarks Track*.

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11), 2600–2606.

Singh, A., D'Arcy, M., Cohan, A., Downey, D., & Feldman, S. (2023). SciRepEval: A multi-format benchmark for scientific document representations. In *Proceedings of EMNLP*.